

Lumina Renewables: what is it worth, and how likely?

Investment committee · fictional company · 50,000 iterations

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BASE CASE (SINGLE POINT) €14.92	SIMULATED MEDIAN €12.49	P5 TO P95 €5.35 to €23.61	VALUE ABOVE €11.50 PRICE 57.5%
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01 Summary

Lumina Renewables S.p.A. is a fictional mid-market Italian solar and wind operator: €145m of revenue, €185m of net debt and 25m shares trading at €11.50. A standard DCF at the most likely value of each assumption gives **€14.92 a share**, 30% above the price. This memo treats six assumptions as probability distributions instead of single points and runs the valuation 50,000 times.

The typical outcome is **€12.49**, not €14.92: the base case sits **19.5% above the simulated median**, because most distributions have their long tail running against value (tax up to 35%, CapEx up to 22%, margin down to 36%). The range is wide: one run in twenty ends below €5.35 and one in twenty above €23.61. At €11.50, 57.5% of runs value the shares above the price. The discount rate alone explains 38% of the spread.

Two checks make the number usable. **Precision:** re-running everything 200 times shows that 2,000 iterations pin the median to ±€0.28 and 50,000 to ±€0.06. **The tax risk:** modelled as an event, a one-in-four chance of a 9-point windfall surcharge, it barely moves the headline, but it splits the outcome in two: a median of €13.29 without the surcharge and €10.20 with it.

CONCLUSION

At €11.50 the shares look modestly cheap on the simulated median (8.6%), not on the base case (29.7%). The edge is thin and rests on two things a buyer should price explicitly: the discount rate, which drives 38% of the variance, and a political decision on energy taxes worth about **€3.09 a share**.

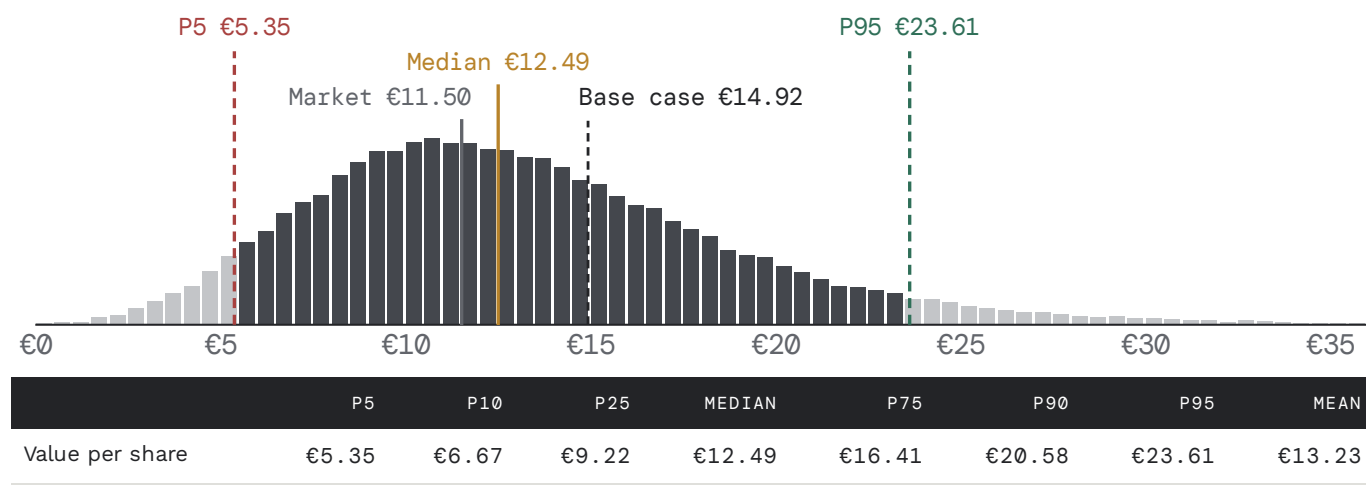
02 Six guesses, six distributions

DRIVER	DISTRIBUTION	MIN / MODE / MAX	WHY THIS SHAPE
Revenue growth (a year)	Triangular	4% / 9% / 14%	Contracted capacity plus new projects
EBITDA margin	Triangular	36% / 42% / 46%	Long tail down: curtailment, lower captured prices
WACC	Normal	8.50% ± 1.10%	Capital-intensive and rate-sensitive
Tax rate	Triangular	24% / 26% / 35%	IRES plus IRAP near 26%; tail for a windfall tax
Terminal growth	Triangular	1% / 2% / 2.75%	Around long-run inflation
CapEx (% of revenue)	Triangular	12% / 16% / 22%	Repowering and grid costs can overrun

Fixed: D&A 13% of revenue, working capital 3% of incremental revenue, five explicit years and a Gordon terminal value. Growth and margin move together (correlation +0.35 through a Gaussian copula). WACC is kept at least 1.5 points above terminal growth and between 5.50% and 12.50%.

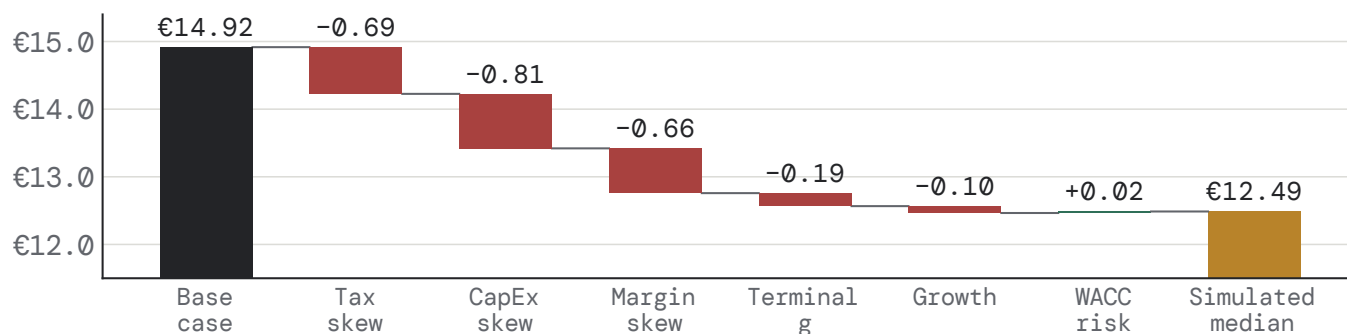
03 A distribution, not a number

Each iteration draws one value for every driver and runs the full DCF. The histogram groups the 50,000 values per share; the darker bars are the middle 90%. The shape is **right-skewed**: a long tail of high outcomes pulls the mean (€13.23) above the median (€12.49). The market price sits just below the median, the base case well above it.



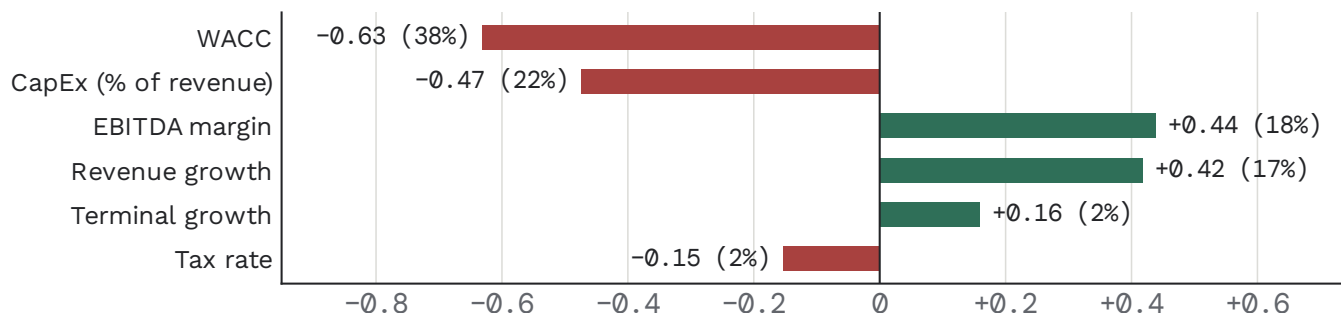
04 Why the base case flatters

The base case plugs in the most likely value of every driver. But a most-likely input does not produce the most-likely output when the distributions are lopsided. Switching the uncertainty on one driver at a time, on the same random numbers, shows where the €2.43 goes: the CapEx tail –€0.81, the tax tail –€0.69, the margin tail –€0.66. WACC risk adds almost nothing to the median (+0.02): it is symmetric, so it widens the range without moving the centre.



05 What moves value

Rank correlation of each driver with value per share across the 50,000 runs; the bracket is each driver's share of the explained variance. **WACC dominates** (38%), followed by CapEx and margin. The tax rate looks minor (2%) because most of the time it barely moves; its risk is concentrated in one event, section 07.



06 How many runs are enough

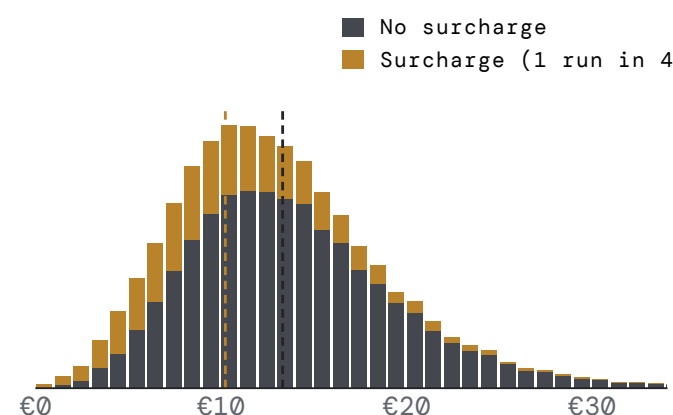
A Monte Carlo result is itself an estimate. To measure how much it moves, the whole simulation was re-run 200 times with fresh random numbers at each size. The table gives the 95% range of each statistic across re-runs. The error falls with the square root of the iterations, and **the tails converge more slowly than the centre**: 2,000 iterations, the size of the Excel workbook, are enough for the median but not for P95.

ITERATIONS	MEDIAN	P5	P95	P(VALUE > PRICE)
1,000	±€0.42	±€0.43	±€1.29	±3.2 pts
2,000	±€0.28	±€0.34	±€0.90	±2.0 pts
10,000	±€0.13	±€0.15	±€0.35	±1.0 pts
50,000	±€0.06	±€0.06	±€0.18	±0.4 pts

The Excel workbook holds the first 2,000 of the 50,000 draws; its median (€12.55) falls inside the 2,000-iteration range, and the Python engine reproduces every one of its 2,000 values to within €0.000000001.

07 The windfall tax as an event

The base model gives the tax rate a long tail up to 35%. In practice a windfall tax is a decision that happens or not. The event model keeps an ordinary rate of 24% to 28% and adds a 9-point surcharge with a 25% probability, so the average tax is the same (28.3% against 28.3%). On the same random numbers, the headline barely moves (median €12.53 against €12.49), but the outcome splits in two:



	NO SURCHARGE	SURCHARGE
Median	€13.29	€10.20
P5	€6.06	€3.65
Value above €11.50	64%	40%

About **€3.09 a share** rides on one decision. A continuous distribution hides that in the tail; the event model makes it something to monitor.

08 Recommendation and limits

- **Anchor on the median, not the base case.** At €11.50 the margin of safety is 8.6%, not 29.7%.
- **Price the two big risks directly.** Stress the discount rate, and track the energy-tax agenda as a binary risk worth about €3.09 a share.
- **Limits.** Lumina is fictional and the ranges are judgement, not market data; the model prices only the risks it defines; low-WACC draws inflate the terminal value, which the 1.5-point guardrail limits but does not remove.

Method and files. Python engine (NumPy, seed 113) in the notebook; Excel workbook with 2,000 live iterations, a windfall sheet and a Python check; live simulator in the browser. Educational project, not investment advice.